Chapter 11

Tearing the Fabric of Space

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f you relentlessly stretch a rubber membrane, sooner or later it will tear. This simple fact has inspired numerous physicists over the years to ask whether the same might be true of the spatial fabric making up the universe. That is, can the fabric of space rip apart, or is this merely a misguided notion that arises from taking the rubber membrane analogy too seriously?

Einstein's general relativity says no, the fabric of space cannot tear.1 The equations of general relativity are firmly rooted in Riemannian geometry and, as we noted in the preceding chapter, this is a framework that analyzes distortions in the distance relations between nearby locations in space. In order to speak meaningfully about these distance relations, the underlying mathematical formalism requires that the substrate of space is smooth—a term with a technical mathematical meaning, but whose everyday usage captures its essence: no creases, no punctures, no separate pieces "stuck" together, and no tears. Were the fabric of space to develop such irregularities, the equations of general relativity would break down, signaling some or other variety of cosmic catastrophe—a disastrous outcome that our apparently well-behaved universe avoids.

This has not kept imaginative theorists over the years from pondering the possibility that a new formulation of physics that goes beyond Einstein's classical theory and incorporates quantum physics might show that rips, tears, and mergers of the spatial fabric can occur. In fact, the realization that quantum physics leads to violent short-distance undulations led some to speculate that rips and tears might be a commonplace microscopic feature of the spatial fabric. The concept of wormholes (a notion with which any fan of Star Trek: Deep Space Nine is familiar) makes use of such musings. The idea is simple: Imagine you're the CEO of a major corporation with headquarters on the ninetieth floor of one of New York City's World Trade Center towers. Through the vagaries of corporate history, an arm of your company with which you need to have ever increasing contact is ensconced on the ninetieth floor of the other tower. As it is impractical